

AdPlatform/DAADD Platform — Complete Technical Guide

For Advertisers, Consumers, Developers, Admins & Merchants

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PLATFORM OVERVIEW

AdPlatform/DAADD is a two-sided AdTech platform where:

- **Advertisers** create campaigns, manage budgets, and track performance across Google Ads, Meta, TikTok, LinkedIn, and Pinterest
- **Consumers** view ads and earn rewards (USD balance)
- **Developers** integrate the platform into their systems via REST API
- **Admins** manage teams, configure system settings, and audit activity
- **Merchants** approve QR code redemptions and manage reward claims

Key Features

- Unified metrics aggregation (5 platforms)
- AI-powered optimization recommendations
- Real-time budget alerts (75%, 90%, 100%)
- A/B testing framework for creatives
- QR code reward redemption (one-time use, HMAC-signed)
- First-party CDP for audience segmentation
- Webhook system for external integrations
- Role-based access control (RBAC)

Base URLs:

- Backend API: <https://api.adplatform.com/api/v1> (prod) or <http://localhost:4000/api/v1> (local)
 - Frontend App: <https://app.adplatform.com> (prod) or <http://localhost:3000> (local)
 - Mobile App: Expo SDK 55 (iOS/Android via EAS Build)
-

ADVERTISER GUIDE

Getting Started as an Advertiser

1. Registration & Authentication

Register a new advertiser account:

```
POST /auth/register
Content-Type: application/json

{
  "email": "advertiser@company.com",
  "password": "SecurePassword123!",
  "name": "John Doe",
  "role": "advertiser"
}
```

Response:

```
{
  "user": {
    "id": "user_abc123",
    "email": "advertiser@company.com",
    "name": "John Doe",
    "role": "advertiser"
  },
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "refreshToken": "refresh_token_xyz789"
}
```

Login:

```
POST /auth/login

{
  "email": "advertiser@company.com",
  "password": "SecurePassword123!"
}
```

Token expires in: 1 hour

Refresh token: Use to get new token without re-entering password

```
POST /auth/refresh
Authorization: Bearer <refresh_token>
```

2. Connect Your Advertising Platforms

Connect accounts from Google Ads, Meta, TikTok, LinkedIn, or Pinterest to sync metrics automatically.

Step 1: Get Authorization URL

```
GET /oauth/authorize/google
Authorization: Bearer <token>
```

Response:

```
{
  "authorization_url": "https://accounts.google.com/o/oauth2/v2/auth?client_id=...",
  "state": "state_token_123"
}
```

Step 2: User Clicks Link & Grants Permission

The platform redirects back to: <https://app.adplatform.com/oauth/callback/google?code=...&state=...>

Step 3: Platform Exchanges Code for Token (automatic)

Our system stores the token **encrypted** (AES-256-GCM) at rest. Tokens refresh automatically every 24 hours.

View Connected Accounts:

```
GET /platform-accounts
Authorization: Bearer <token>
```

Response:

```
{
  "data": [
    {
      "id": "pa_google_001",
      "platform": "google",
      "platform_account_id": "123-456-789",
      "platform_account_name": "My Google Ads Account",
      "status": "connected",
      "is_active": true,
      "sync_frequency": "realtime",
      "last_synced": "2026-05-17T14:32:00Z",
      "created_at": "2026-05-15T10:00:00Z"
    }
  ]
}
```

Statuses:

- connected — Token valid, syncing metrics
- pending — OAuth in progress
- sync_error — Last sync failed; check error_message
- revoked — User revoked access on platform; re-authorize needed

3. Create Your First Campaign

Create a new campaign:

```
POST /campaigns
Authorization: Bearer <token>
Content-Type: application/json

{
  "name": "Summer Sale Campaign 2026",
  "description": "Drive traffic to summer product launch",
  "industry": "Retail",
  "start_date": "2026-06-01T00:00:00Z",
  "end_date": "2026-08-31T23:59:59Z",
  "budget_total": 5000.00,
  "reward_value": 2.50,
  "is_age_restricted": false,
  "enable_ai_optimization": true,
  "language": "en",
  "platform_ids": ["pa_google_001", "pa_meta_001"],
  "targeting_config": {
    "regions": ["US", "CA"],
    "devices": ["desktop", "mobile"],
    "languages": ["en"]
  }
}
```

Response:

```
{
  "id": "camp_summer_2026",
  "name": "Summer Sale Campaign 2026",
  "status": "DRAFT",
  "advertiser_id": "user_abc123",
  "budget_total": 5000.00,
  "budget_spent": 0,
  "created_at": "2026-05-17T15:00:00Z"
}
```

Campaign Status Machine:

- DRAFT → (click "Launch") → ACTIVE → (end date/budget) → COMPLETED or PAUSED

4. Upload Creative Assets

Add images/videos to your campaign:

```
POST /campaigns/camp_summer_2026/creatives
Authorization: Bearer <token>
Content-Type: multipart/form-data

file: <image.jpg or video.mp4>
```

Supported formats:

- Images: JPG, PNG, WebP (max 10MB)
- Videos: MP4, WebM, MOV (max 100MB)

Response:

```
{
  "id": "creative_img_001",
  "campaign_id": "camp_summer_2026",
  "type": "image",
  "url": "https://cdn.adplatform.com/creatives/img_001.jpg",
  "format": "jpg",
  "file_size": 2048000
}
```

5. Monitor Campaign Performance

Unified Metrics Dashboard

```
GET /dashboard/unified-metrics?startDate=2026-06-01&endDate=2026-06-30
Authorization: Bearer <token>
```

Response:

```
{
  "total_metrics": {
    "impressions": 125000,
    "clicks": 3750,
    "conversions": 187,
    "spend": 1250.00,
    "ctr": 3.0,
    "cpc": 0.33,
    "cpa": 6.68,
    "roas": 1.5
  },
  "by_platform": {
    "google": { "impressions": 75000, "clicks": 2250, ... },
    "meta": { "impressions": 50000, "clicks": 1500, ... }
  },
  "daily_breakdown": [
    { "date": "2026-06-01", "impressions": 4000, "clicks": 120, ... },
    ...
  ]
}
```

Campaign-Specific Metrics

```
GET /campaigns/camp_summer_2026/analytics
Authorization: Bearer <token>
```

6. Receive Budget Alerts

When your campaign reaches spending thresholds, you'll receive:

1. **75% spent** → Email alert
2. **90% spent** → Email + in-app notification

3. **100% spent** → Campaign auto-paused + notification

Check notifications in-app:

```
GET /notifications
Authorization: Bearer <token>

# Mark as read:
PATCH /notifications/notif_001
{ "is_read": true }
```

7. Enable AI Optimization

Get automatic recommendations on budget allocation, creative optimization, and scaling strategies.

Get AI recommendations:

```
GET /ai/recommendations?campaignId=camp_summer_2026
Authorization: Bearer <token>
```

Response:

```
{
  "recommendations": [
    {
      "id": "rec_001",
      "type": "scale_winning_platform",
      "campaign_id": "camp_summer_2026",
      "title": "Scale Google Ads (High ROAS)",
      "description": "Google Ads has 3x ROAS. Increase budget by $500.",
      "impact_potential": 85,
      "action": {
        "type": "increase_budget",
        "platform": "google",
        "amount": 500
      }
    }
  ]
}
```

Apply a recommendation:

```
POST /ai/apply/rec_001
Authorization: Bearer <token>

# Recommendation is auto-applied and logged in audit trail
```

8. A/B Test Your Creatives

Create A/B test:

POST /campaigns/camp_summer_2026/ab-test/create

Authorization: Bearer <token>

Content-Type: application/json

```
{
  "campaignId": "camp_summer_2026",
  "controlCreativeId": "creative_img_001",
  "variantCreativeIds": ["creative_img_002", "creative_img_003"],
  "trafficAllocation": 20
}
```

Get test results:

GET /campaigns/camp_summer_2026/ab-test/results

Authorization: Bearer <token>

Response:

```
{
  "testId": "test_ab_001",
  "campaignId": "camp_summer_2026",
  "controlCreative": {
    "id": "creative_img_001",
    "metrics": {
      "impressions": 10000,
      "clicks": 300,
      "ctr": 3.0
    }
  },
  "variantCreatives": [
    {
      "id": "creative_img_002",
      "metrics": {
        "impressions": 10000,
        "clicks": 400,
        "ctr": 4.0
      }
    }
  ],
  "winner": {
    "creativeId": "creative_img_002",
    "ctr": 4.0
  }
}
```

9. Clone & Reuse Successful Campaigns

Clone a campaign:

POST /campaigns/camp_summer_2026/clone

Authorization: Bearer <token>

Response:

```
{
  "id": "camp_summer_2026_copy",
  "name": "Summer Sale Campaign 2026 (Copy)",
  "status": "DRAFT",
  "budget_total": 5000.00,
  "budget_spent": 0,
  "creatives": [
    // All creatives from original copied
  ]
}
```

10. Issue QR Code Rewards

Allow consumers to scan QR codes and claim rewards.

Generate QR code:

```
POST /redemption/generate-qr
Authorization: Bearer <token>
Content-Type: application/json

{
  "campaign_id": "camp_summer_2026",
  "amount": 2.50,
  "quantity": 100
}
```

Response:

```
{
  "qr_codes": [
    {
      "id": "qr_001",
      "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
      "qr_image_url": "https://cdn.adplatform.com/qr/qr_001.png",
      "amount": 2.50,
      "is_redeemed": false,
      "expires_at": "2026-05-17T16:02:00Z",
      "created_at": "2026-05-17T16:00:00Z"
    }
  ]
}
```

QR Code Security:

- HMAC-SHA256 signed tokens (cannot be forged)
- 2-minute expiry (single-use)
- Consumer scans → Automatic balance credit

CONSUMER/USER GUIDE

Getting Started as a Consumer

1. Registration & Mobile App Setup

Download the mobile app via Expo or app store, then register:

Register:

```
POST /auth/register
{
  "email": "consumer@example.com",
  "password": "SecurePassword123!",
  "name": "Alice Johnson",
  "role": "consumer"
}
```

Login:

```
POST /auth/login
{
  "email": "consumer@example.com",
  "password": "SecurePassword123!"
}
```

Store token securely:

- Mobile: expo-secure-store (encrypted keychain)
- Web: httpOnly cookie (server-set)

2. Browse Available Ads

Get featured ads:

```
GET /ads?featured=true
(No auth required — public endpoint)
```

Response:

```
{
  "data": [
    {
      "id": "ad_summer_001",
      "advertiser": {
        "id": "advertiser_123",
        "name": "Summer Brand Co.",
        "logo": "https://cdn.adplatform.com/logos/summer_brand.png"
      },
      "title": "Summer Collection 2026",
      "description": "Shop our new summer line",
      "creative_url": "https://cdn.adplatform.com/creatives/summer.jpg",
      "reward_amount": 2.50,
      "is_age_restricted": false,
      "category": "Retail",
      "ctr": 3.5,
      "average_rating": 4.2,
      "review_count": 156
    }
  ]
}
```

3. View Ad Details & Reviews

Get ad details:

```
GET /ads/ad_summer_001
```

Response:

```
{
  "id": "ad_summer_001",
  "title": "Summer Collection 2026",
  "reward_amount": 2.50,
  "reviews": [
    {
      "id": "review_001",
      "user_name": "Bob Smith",
      "rating": 5,
      "comment": "Great quality products!",
      "created_at": "2026-05-10T12:00:00Z"
    }
  ],
  "rating_summary": {
    "average": 4.2,
    "total_reviews": 156,
    "distribution": {
      "5_star": 120,
      "4_star": 25,
      "3_star": 8,
      "2_star": 2,
      "1_star": 1
    }
  }
}
```

4. View Ads & Earn Impressions

Track ad view (impression):

```
POST /events/track
Authorization: Bearer <token>
Content-Type: application/json

{
  "event_type": "IMPRESSION",
  "campaign_id": "camp_summer_2026",
  "ad_id": "ad_summer_001",
  "user_id": "user_consumer_123"
}
```

Track viewable impression (ad visible for 1+ second):

```
POST /events/track
{
  "event_type": "VIEWABLE_IMPRESSION",
  "campaign_id": "camp_summer_2026",
  "ad_id": "ad_summer_001",
  "user_id": "user_consumer_123"
}
```

5. Click & Convert

Track click:

```
POST /events/track
{
  "event_type": "CLICK",
  "campaign_id": "camp_summer_2026",
  "ad_id": "ad_summer_001",
  "user_id": "user_consumer_123"
}
```

Track conversion (purchase/action on advertiser site):

```
POST /events/track
{
  "event_type": "CONVERSION",
  "campaign_id": "camp_summer_2026",
  "ad_id": "ad_summer_001",
  "user_id": "user_consumer_123",
  "metadata": {
    "conversion_value": 50.00,
    "purchase_id": "order_12345"
  }
}
```

6. Claim Rewards via QR Code

Scan QR code with mobile app:

The app extracts the token from the QR code and sends:

```
POST /rewards/claim
Authorization: Bearer <token>
Content-Type: application/json

{
  "qr_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

Response:

```
{
  "success": true,
  "reward_amount": 2.50,
  "new_balance": 25.50,
  "message": "Reward claimed! Check your balance in the app."
}
```

Error cases:

- "Invalid or expired QR code" — Token expired (2 min) or already redeemed
- "Already claimed" — QR code was used before (one-time only)
- "Invalid token" — QR was tampered with (HMAC mismatch)

7. Check Your Reward Balance

Get reward balance:

```
GET /rewards/balance
Authorization: Bearer <token>
```

Response:

```
{
  "total_balance": 25.50,
  "currency": "USD",
  "recent_claims": [
    {
      "id": "claim_001",
      "campaign_id": "camp_summer_2026",
      "amount": 2.50,
      "claimed_at": "2026-05-17T14:00:00Z"
    }
  ]
}
```

8. Write Reviews

Submit a review for an ad:

```
POST /reviews
Authorization: Bearer <token>
Content-Type: application/json

{
  "campaign_id": "camp_summer_2026",
  "rating": 5,
  "comment": "Amazing product quality! Would buy again."
}
```

Response:

```
{
  "id": "review_user_001",
  "campaign_id": "camp_summer_2026",
  "user_id": "user_consumer_123",
  "rating": 5,
  "comment": "Amazing product quality! Would buy again.",
  "created_at": "2026-05-17T15:30:00Z"
}
```

Constraints:

- One review per campaign per user
- Edit/delete not allowed (immutable)
- Only eligible if you've claimed a reward from that campaign

9. Age Verification for Restricted Content

Some ads are age-restricted. You'll be prompted to verify.

Request OTP:

```
POST /auth/age-verify/request
Authorization: Bearer <token>
```

Response:

```
{
  "expires_in": 600,
  "message": "OTP sent to your registered email"
}
```

Verify OTP:

```
POST /auth/age-verify/confirm
Authorization: Bearer <token>
{
  "otp": "123456"
}
```

10. Manage Your Profile

Get profile:

```
GET /users/me
Authorization: Bearer <token>
```

Update profile:

```
PATCH /users/me
Authorization: Bearer <token>
{
  "name": "Alice Johnson",
  "email": "alice.new@example.com"
}
```

DEVELOPER GUIDE

API Integration & Authentication

1. Authentication Flow

All API endpoints (except public ones like /ads, /pixel) require:

```
Authorization: Bearer <JWT_TOKEN>
```

JWT Token Structure:

```
{
  "sub": "user_abc123",
  "email": "developer@example.com",
  "role": "advertiser",
  "iat": 1716000000,
  "exp": 1716003600
}
```

Token Expiry: 1 hour

Refresh: Use /auth/refresh endpoint

2. Webhook Integration

Set up real-time event notifications to your HTTPS endpoint.

Register webhook:

```
POST /webhooks
Authorization: Bearer <token>
Content-Type: application/json

{
  "url": "https://your-api.example.com/webhooks/adplatform",
  "events": [
    "campaign.anomaly_detected",
    "campaign.budget_threshold",
    "campaign.completed",
    "conversion.received"
  ]
}
```

Response:

```
{
  "id": "webhook_001",
  "url": "https://your-api.example.com/webhooks/adplatform",
  "secret": "whsec_abc123xyz789...",
  "events": ["campaign.anomaly_detected", ...],
  "is_active": true
}
```

Receive webhook payload:

```
{
  "event": "campaign.budget_threshold",
  "timestamp": "2026-05-17T14:30:00Z",
  "data": {
    "campaign_id": "camp_summer_2026",
    "campaign_name": "Summer Sale Campaign 2026",
    "threshold": 75,
    "spent": 3750.00,
    "budget": 5000.00
  },
  "signature": "t=1716000000,v1=1234567890abcdef..."
}
```

Verify webhook signature (HMAC-SHA256):

```
const crypto = require('crypto');
const secret = 'whsec_abc123xyz789...';
const [timestamp, signature] = req.headers['x-adplatform-signature'].split(',');

const signed = crypto
  .createHmac('sha256', secret)
  .update(`${timestamp}.${JSON.stringify(req.body)}`)
  .digest('hex');

if (signed !== signature) {
  throw new Error('Invalid webhook signature');
}
```

3. Conversion Pixel Integration

Use for server-to-server conversion tracking (affiliate/partner sites).

Post conversion:

```
POST /pixel/camp_summer_2026?uid=user_123&ev=conversion&val=50&ref=affiliate.com
(No auth required — public endpoint)
```

Parameters:

- uid (optional) — User ID for cross-device tracking
- ev (required) — Event type (only "conversion" supported)
- val (optional) — Conversion value in cents (e.g., 50 = \$0.50)
- ref (optional) — Referrer domain

Response: 1x1 transparent GIF (standard pixel format)

Rate limits: 100 requests/minute per campaign

4. Analytics API

Get unified metrics:

```
GET /dashboard/unified-metrics?startDate=2026-06-01&endDate=2026-06-30
Authorization: Bearer <token>
```


Get campaign breakdown:

```
GET /analytics/dashboard?campaignId=camp_summer_2026
Authorization: Bearer <token>
```

Get creative A/B test metrics:

```
GET /analytics/creatives/camp_summer_2026
Authorization: Bearer <token>
```

Export to CSV:

```
GET /analytics/export/csv?format=csv&campaignId=camp_summer_2026
Authorization: Bearer <token>
```

Export to PDF:

```
GET /analytics/export/pdf?format=pdf&campaignId=camp_summer_2026
Authorization: Bearer <token>
```

5. OAuth2 Integration (for platforms)

Connect to Google Ads, Meta, TikTok, LinkedIn, or Pinterest.

Get authorization URL:

```
GET /oauth/authorize/{platform}
Authorization: Bearer <token>
```

Supported platforms: google, meta, tiktok, linkedin, pinterest

Platform redirects back with code:

```
https://app.adplatform.com/oauth/callback/{platform}?code=...&state=...
```

Our system exchanges code for token (automatic):

- Token stored encrypted (AES-256-GCM)
- Auto-refreshes 24 hours before expiry

6. Error Handling

All errors follow this format:

```
{
  "error": {
    "message": "Campaign not found",
    "code": "NOT_FOUND",
    "status": 404
  }
}
```

Common error codes:

- INVALID_INPUT (400) — Validation failed
- UNAUTHORIZED (401) — Missing/invalid token

- FORBIDDEN (403) — Insufficient permissions
- NOT_FOUND (404) — Resource doesn't exist
- CONFLICT (409) — Resource state conflict
- RATE_LIMITED (429) — Too many requests
- INTERNAL_ERROR (500) — Server error

7. Pagination

List endpoints support pagination:

```
GET /campaigns?page=1&limit=20&sort=created_at&order=desc
```

Response:

```
{
  "data": [...],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 150,
    "totalPages": 8,
    "hasNextPage": true,
    "hasPreviousPage": false
  }
}
```

8. Rate Limiting

API rate limits:

- **Standard endpoints:** 1000 req/hour per API key
- **Pixel endpoint:** 100 req/minute per campaign
- **Webhook delivery:** 3 retry attempts with exponential backoff

Rate limit headers:

```
X-RateLimit-Limit: 1000
X-RateLimit-Remaining: 999
X-RateLimit-Reset: 1716003600
```

ADMIN GUIDE

System Administration

1. User Management

List all users:

```
GET /users
Authorization: Bearer <admin_token>
```

Response:

```
{
  "data": [
    {
      "id": "user_abc123",
      "email": "advertiser@company.com",
      "name": "John Doe",
      "role": "advertiser",
      "created_at": "2026-05-15T10:00:00Z",
      "last_login": "2026-05-17T14:00:00Z"
    }
  ]
}
```

Create user (admin-only):

```
POST /users
Authorization: Bearer <admin_token>
{
  "email": "newuser@company.com",
  "password": "TempPassword123!",
  "name": "Jane Smith",
  "role": "campaign_manager"
}
```

Update user role:

```
PATCH /users/user_abc123
Authorization: Bearer <admin_token>
{
  "role": "admin"
}
```

Deactivate user:

```
DELETE /users/user_abc123
Authorization: Bearer <admin_token>
```

2. Team Management

Create team:

```
POST /teams
Authorization: Bearer <admin_token>
{
  "name": "Marketing Team",
  "description": "Handles all campaign management"
}
```

Invite team member:

```
POST /teams/team_001/invite
Authorization: Bearer <admin_token>
{
  "email": "team_member@company.com",
  "role": "campaign_manager"
}
```

Role-based permissions:

- admin — Full system access, user management, team configuration
- campaign_manager — Create/edit campaigns, view analytics
- viewer — View-only access to dashboards

3. Audit Logging

Get audit logs:

```
GET /audit-logs?type=campaign_created&startDate=2026-05-01
Authorization: Bearer <admin_token>
```

Response:

```
{
  "data": [
    {
      "id": "audit_001",
      "action": "campaign_created",
      "user_id": "user_abc123",
      "resource_id": "camp_summer_2026",
      "timestamp": "2026-05-17T15:00:00Z",
      "details": {
        "campaign_name": "Summer Sale Campaign 2026",
        "budget": 5000.00
      }
    }
  ]
}
```

Audit event types:

- campaign_created, campaign_updated, campaign_deleted
- oauth_connected, oauth_revoked
- user_invited, user_role_changed
- ai_recommendation_applied
- anomaly_detected

4. System Configuration

Update system settings:

```
PATCH /system/settings
Authorization: Bearer <admin_token>
{
  "cors_origins": ["https://app.adplatform.com", "https://staging.adplatform.com"],
  "email_provider": "resend",
  "max_campaign_budget": 100000.00,
  "otp_expiry_minutes": 10
}
```

Check system health:

```
GET /system/health
(No auth required)
```

Response:

```
{
  "status": "healthy",
  "timestamp": "2026-05-17T15:00:00Z",
  "services": {
    "api": "operational",
    "mongodb": "operational",
    "redis": "operational",
    "email_service": "operational",
    "oauth_providers": {
      "google": "operational",
      "meta": "operational"
    }
  }
}
```

5. Analytics & Reporting

Get platform-wide metrics:

```
GET /admin/metrics?startDate=2026-05-01&endDate=2026-05-17
Authorization: Bearer <admin_token>
```

Response:

```
{
  "total_advertisers": 234,
  "total_consumers": 12450,
  "active_campaigns": 89,
  "total_spend": 125000.00,
  "total_rewards_issued": 15000.00,
  "total_impressions": 5000000,
  "total_conversions": 50000
}
```

MERCHANT GUIDE

Managing Reward Redemptions

1. Approve/Reject QR Code Claims

Merchants review and approve reward claims from consumers.

Get pending redemptions:

```
GET /redemption/pending
Authorization: Bearer <merchant_token>
```

Response:

```
{
  "data": [
    {
      "id": "redemption_001",
      "qr_code_id": "qr_001",
      "campaign_id": "camp_summer_2026",
      "consumer_id": "user_consumer_123",
      "amount": 2.50,
      "status": "pending",
      "claimed_at": "2026-05-17T14:30:00Z"
    }
  ]
}
```

Approve redemption:

```
POST /redemption/approve
Authorization: Bearer <merchant_token>
{
  "redemption_id": "redemption_001",
  "action": "approve"
}
```

Reject redemption:

```
POST /redemption/approve
Authorization: Bearer <merchant_token>
{
  "redemption_id": "redemption_001",
  "action": "reject",
  "reason": "Invalid purchase proof"
}
```

2. Dynamic Redemption Control

Control how rewards are redeemed in real-time.

Set redemption rules:

```
POST /merchant/redemption-control
Authorization: Bearer <merchant_token>
{
  "campaign_id": "camp_summer_2026",
  "min_purchase_amount": 50.00,
  "max_daily_redemptions": 100,
  "require_purchase_proof": true,
  "proof_submission_method": "email"
}
```

Response:

```
{
  "id": "control_001",
  "campaign_id": "camp_summer_2026",
  "is_active": true,
  "created_at": "2026-05-17T10:00:00Z"
}
```

3. Redemption History & Reporting

Get redemption history:

```
GET /merchant/redemptions?status=approved&startDate=2026-05-01
Authorization: Bearer <merchant_token>
```

Response:

```
{
  "data": [
    {
      "id": "redemption_001",
      "consumer_name": "Alice Johnson",
      "amount": 2.50,
      "status": "approved",
      "approved_at": "2026-05-17T14:45:00Z"
    }
  ],
  "summary": {
    "total_approved": 500,
    "total_rejected": 10,
    "total_amount_approved": 1250.00
  }
}
```

⚙️ COMMON TASKS

Task: Export Campaign Analytics as PDF

```
// Backend example
const campaignId = 'camp_summer_2026';
const response = await fetch(
  `https://api.adplatform.com/api/v1/analytics/export/pdf?campaignId=${campaignId}`,
  {
    headers: {
      'Authorization': `Bearer ${token}`,
      'Content-Type': 'application/pdf'
    }
  }
);
const pdfBuffer = await response.arrayBuffer();

// Save or send via email
```

Task: Integrate Conversion Pixel into E-Commerce Site

```
<!-- On checkout success page -->


<!-- Alternative: JSON POST via JavaScript -->
<script>
  const campaignId = 'camp_summer_2026';
  const userId = getCookie('adplatform_user_id'); // Set when user clicks ad
  const orderValue = 50.00 * 100; // Convert to cents

  fetch(`https://api.adplatform.com/api/v1/pixel/${campaignId}?
uid=${userId}&ev=conversion&val=${orderValue}`, {
    method: 'POST'
  });
</script>
```

Task: Automate Daily Campaign Reports via Webhook

```
// Your backend receives webhook
app.post('/webhooks/adplatform', (req, res) => {
  const { event, data } = req.body;

  if (event === 'campaign.completed') {
    // Send final report email
    sendEmail(advertiserEmail, `Campaign ${data.campaign_name} completed`, {
      total_spend: data.total_spend,
      total_conversions: data.total_conversions,
      roi: data.roas
    });
  }

  res.json({ received: true });
});
```


Task: Set Up Automated Budget Pacing Alerts

```
// Webhook receives budget threshold events automatically
// You can then integrate with Slack, PagerDuty, etc.

app.post('/webhooks/adplatform', async (req, res) => {
  const { event, data } = req.body;

  if (event === 'campaign.budget_threshold') {
    const slackMessage = {
      text: `Budget Alert: ${data.campaign_name} is ${data.threshold}% spent`,
      blocks: [{
        type: "section",
        text: {
          type: "mrkdwn",
          text: `    *${data.campaign_name}*\n` +
            `    Spent: ${data.spent.toFixed(2)} / ${data.budget.toFixed(2)}`
        }
      ]
    };

    await fetch(process.env.SLACK_WEBHOOK_URL, {
      method: 'POST',
      body: JSON.stringify(slackMessage)
    });
  }

  res.json({ received: true });
});
```

API REFERENCE

Authentication Endpoints

Method	Endpoint	Auth	Purpose
POST	/auth/register	None	Create new account
POST	/auth/login	None	Login and get token
POST	/auth/refresh	Refresh Token	Get new access token
PATCH	/auth/change-password	Bearer	Change password
POST	/auth/forgot-password	None	Request password reset
POST	/auth/reset-password	None	Reset password with token
GET	/auth/me	Bearer	Get current user

Campaign Management

Method	Endpoint	Auth	Purpose
GET	/campaigns	Bearer	List campaigns

GET	/campaigns/:id	Bearer	Get campaign details
POST	/campaigns	Bearer	Create campaign
PATCH	/campaigns/:id	Bearer	Update campaign
DELETE	/campaigns/:id	Bearer	Delete campaign
POST	/campaigns/:id/clone	Bearer	Clone campaign
PATCH	/campaigns/:id/status	Bearer	Update status

Platform Accounts

Method	Endpoint	Auth	Purpose
GET	/platform-accounts	Bearer	List connected accounts
POST	/oauth/authorize/:platform	Bearer	Get OAuth authorization URL
GET	/oauth/callback/:platform	None	OAuth callback (auto-handled)
PATCH	/platform-accounts/:id	Bearer	Update account settings
DELETE	/platform-accounts/:id	Bearer	Disconnect account

Analytics & Reporting

Method	Endpoint	Auth	Purpose
GET	/dashboard/unified-metrics	Bearer	Get unified metrics
GET	/analytics/dashboard	Bearer	Campaign breakdown
GET	/analytics/creatives/:campaignId	Bearer	A/B test metrics
GET	/analytics/export/csv	Bearer	Export CSV
GET	/analytics/export/pdf	Bearer	Export PDF

AI Optimization

Method	Endpoint	Auth	Purpose
GET	/ai/recommendations	Bearer	Get AI recommendations
POST	/ai/apply/:recommendationId	Bearer	Apply recommendation

Rewards & Redemption

Method	Endpoint	Auth	Purpose
POST	/rewards/claim	Bearer	Claim reward via QR
GET	/rewards/balance	Bearer	Get reward balance
POST	/redemption/generate-qr	Bearer	Generate QR codes
POST	/redemption/approve	Bearer	Approve/reject claims

Webhooks

Method	Endpoint	Auth	Purpose
POST	/webhooks	Bearer	Register webhook
GET	/webhooks	Bearer	List webhooks
PATCH	/webhooks/:id	Bearer	Update webhook
DELETE	/webhooks/:id	Bearer	Delete webhook

Public Endpoints (No Auth)

Method	Endpoint	Purpose
GET	/ads	Browse ads
GET	/ads/:id	Get ad details
POST	/pixel/:campaignId	Server-to-server conversion tracking
GET	/system/health	System health status

TROUBLESHOOTING

Common Issues & Solutions

"Invalid or expired token"

Cause: JWT token expired (1 hour) or malformed

Solution: Call /auth/refresh with refresh token to get new access token

"Campaign not found"

Cause: Campaign ID doesn't exist or belongs to another user

Solution: Double-check campaign ID; verify you're accessing your own campaigns

"Budget alert not received"

Cause: Email service down or address marked as spam

Solution: Check email spam folder; verify email address in account settings

"QR code invalid or expired"

Cause: Code expired (2 min) or was already redeemed

Solution: Generate new QR code; each code is single-use

"Webhook not receiving events"

Cause: URL unreachable, signature verification failed, or wrong events subscribed

Solution: Verify HTTPS endpoint is public; check webhook logs; verify secret

"OAuth token refresh failed"

Cause: Platform revoked access; user changed password

Solution: Re-authorize the platform via /oauth/authorize/{platform}

"Rate limit exceeded"

Cause: Too many requests in short time

Solution: Implement exponential backoff; check rate limit headers

"Conversion not attributed to campaign"

Cause: Pixel fired after attribution window (default 30 days)

Solution: Adjust attribution window via /attribution/window/:campaignId

Support & Resources

- **API Documentation:** <https://api.adplatform.com/api/docs>
 - **Status Page:** <https://status.adplatform.com>
 - **Support Email:** support@adplatform.com
 - **Slack Channel:** #adplatform-developers (for team members)
 - **GitHub:** <https://github.com/adplatform/platform> (open-source components)
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